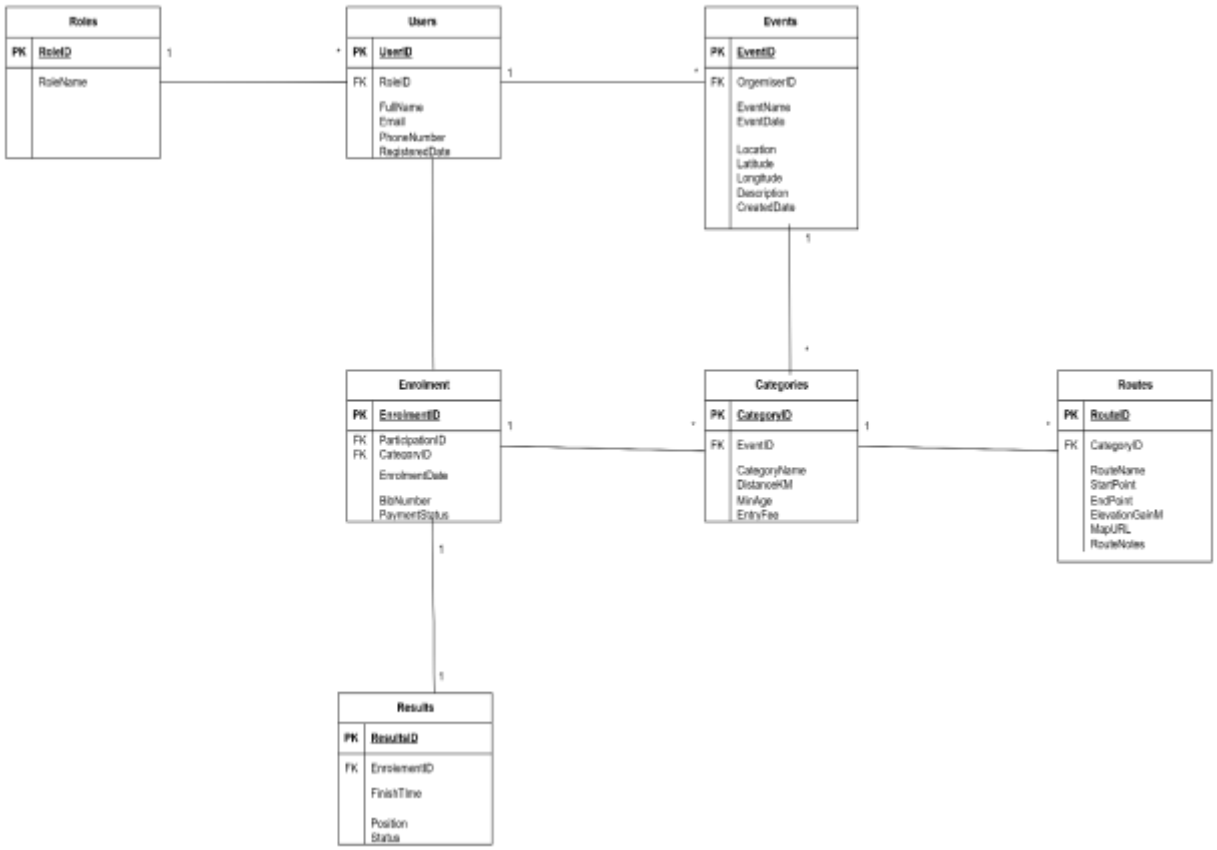
NAME: DESIRE MORABA

STUDENT NUMBER: ST10491510

MODULE CODE: PROG6212

PART 1



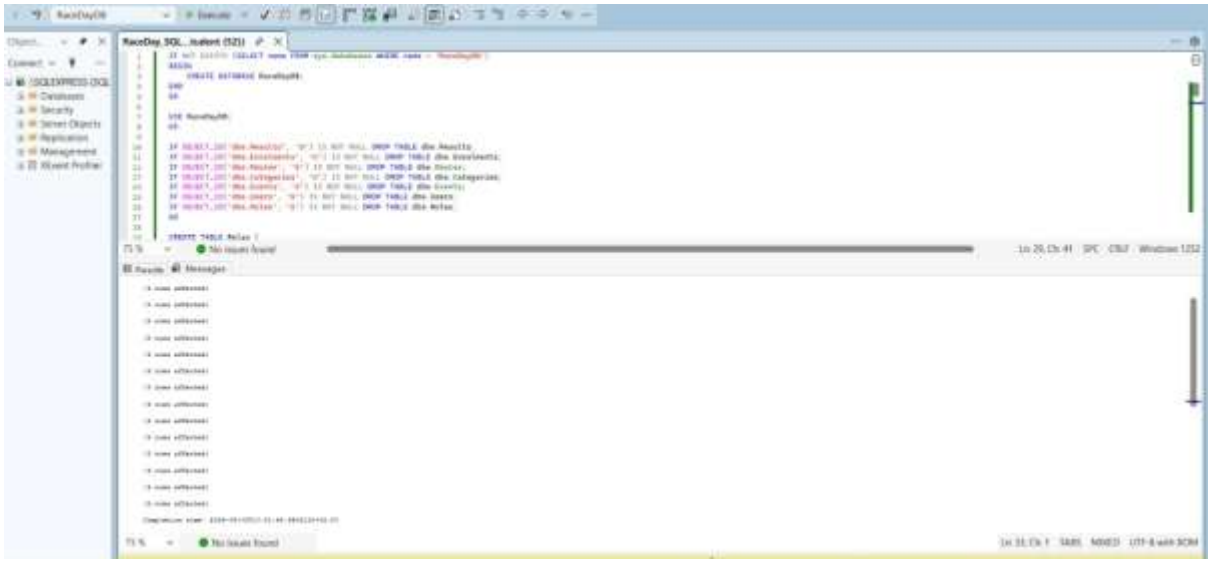
This ERD represents a Race Day registration and event management system. It has seven entities that define different aspects of the system.

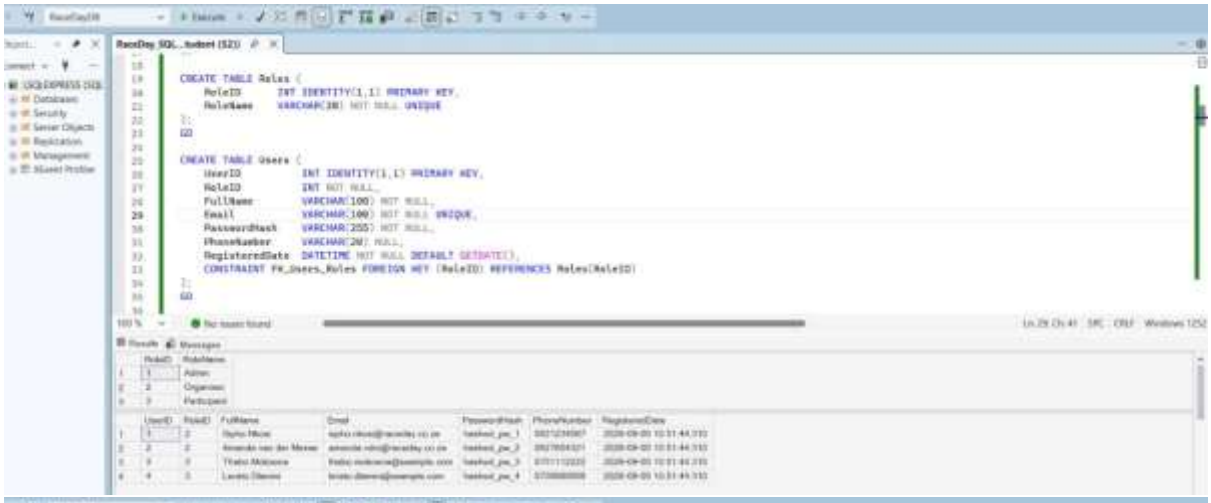
SECTION B

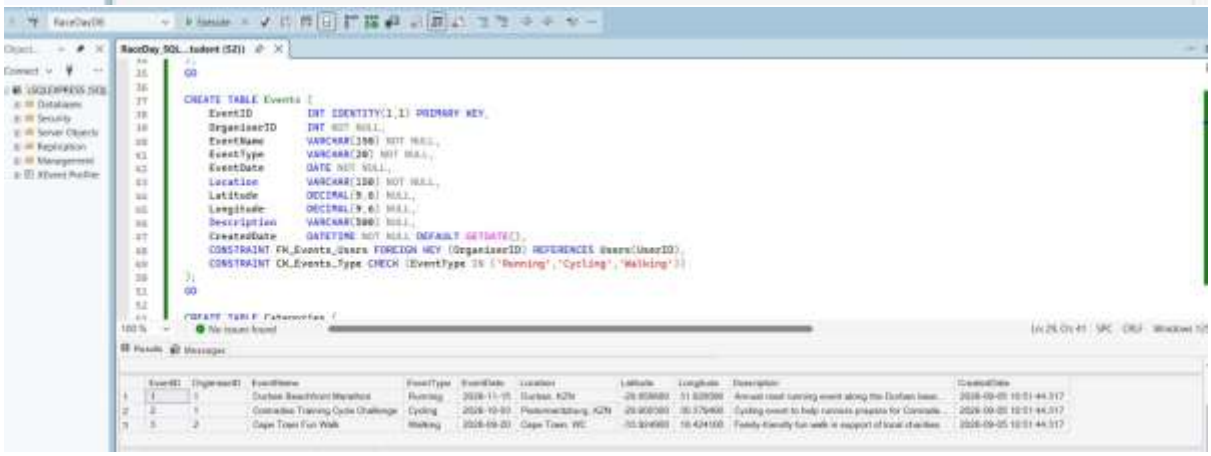
RaceDay – API Endpoint Plan

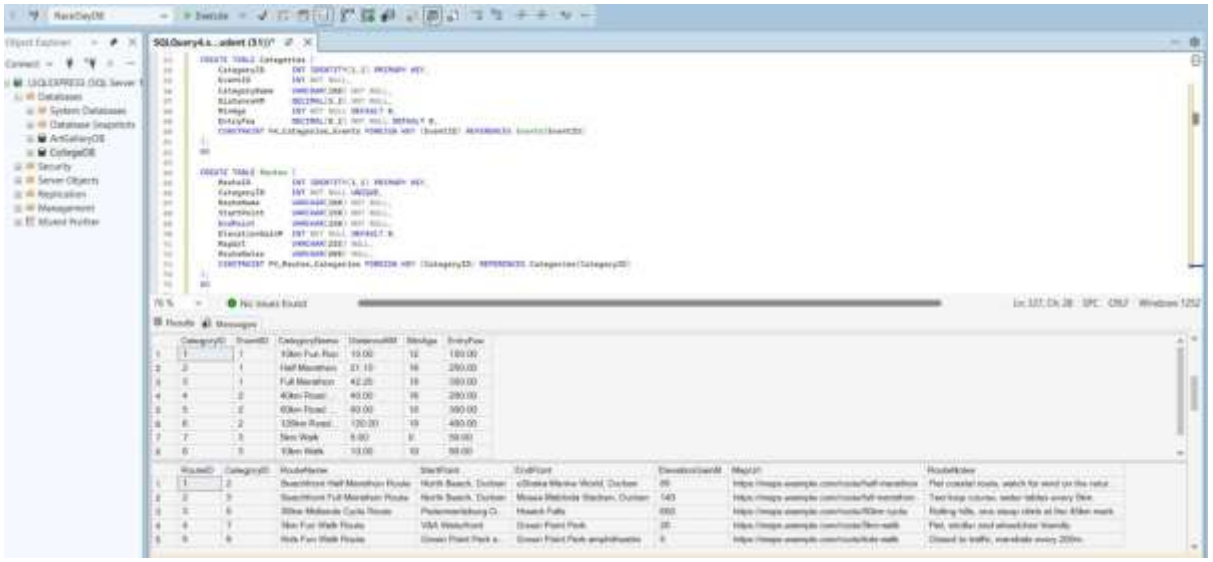
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either an Organiser or a Participant.	None (public)	{ fullName, email, password, phoneNumber, role }	201 Created - user record created; 400 Bad Request invalid/missing fields; 409 Conflict - email already registered
POST	/api/auth/login	Authenticates a user and returns an access token.	None (public)	{ email, password }	200 OK - token + user info; 401 Unauthorized - invalid credentials
GET	/api/users/me	Retrieves the logged-in user's own profile.	Any (logged in)	None	200 OK - user profile; 401 Unauthorized - not logged in
PUT	/api/users/me	Updates the logged-in user's own profile details.	Any (logged in)	{ fullName, phoneNumber }	200 OK - updated profile; 400 Bad Request - invalid fields
GET	/api/events	Lists all upcoming events, optionally filtered by type or date.	None (public)	None	200 OK - array of events
GET	/api/events/{id}	Retrieves full details of a single event, including its categories.	None (public)	None	200 OK - event details; 404 Not Found - event does not exist
POST	/api/events	Creates a new event.	Organiser	{ eventName, eventType, eventDate, location, latitude, longitude, description }	201 Created - new event; 400 Bad Request - invalid fields
PUT	/api/events/{id}	Updates an event owned by the logged-in organiser.	Organiser	{ eventName, eventType, eventDate, location, latitude, longitude, description }	200 OK - updated event; 403 Forbidden - not the owning organiser; 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in organiser.	Organiser	None	204 No Content; 403 Forbidden - not the owning organiser; 404 Not Found
GET	/api/events/{id}/weather	Fetches a live weather forecast for the event's location and date from an external weather API, so participants can prepare for race day.	None (public)	None	200 OK - forecast (temperature, wind, rain chance); 404 Not Found - event does not exist; 502 Bad Gateway weather provider unavailable
GET	/api/events/{id}/categories	Lists all categories belonging to a specific event.	None (public)	None	200 OK - array of categories; 404 Not Found - event does not exist
POST	/api/events/{id}/categories	Adds a new category to an event.	Organiser	{ categoryName, distanceKM, minAge, entryFee }	201 Created - new category; 403 Forbidden; 404 Not Found
PUT	/api/categories/{id}	Updates an existing category.	Organiser	{ categoryName, distanceKM, minAge, entryFee }	200 OK - updated category; 403 Forbidden; 404 Not Found
DELETE	/api/categories/{id}	Deletes a category from an event.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found
GET	/api/categories/{id}/route	Retrieves the route details (start/end point, elevation, map link) for a category, so participants can preview the course.	None (public)	None	200 OK - route details; 404 Not Found - no route set for this category
POST	/api/categories/{id}/route	Creates or replaces the route info for a category.	Organiser	{ routeName, startPoint, endPoint, elevationGainM, mapUrl, routeNotes }	201 Created - route saved; 403 Forbidden; 404 Not Found - category does not exist
POST	/api/categories/{id}/enrol	Enrols the logged-in participant into a category and generates a bib number.	Participant	{ } (participant identified from login token)	201 Created - enrolment record with bib number; 404 Not Found - category does not exist; 409 Conflict - already enrolled
GET	/api/users/me/enrolments	Lists all events/categories the logged-in participant is currently enrolled in.	Participant	None	200 OK - array of enrolments
DELETE	/api/enrolments/{id}	Cancels an enrolment belonging to the logged-in participant.	Participant	None	204 No Content; 403 Forbidden - not the owning participant; 404 Not Found
GET	/api/users/me/history	Retrieves the logged-in participant's personal performance history - every past event, category, finish time, and position.	Participant	None	200 OK - array of past results ordered by event date
GET	/api/categories/{id}/results	Retrieves the results/leaderboard for a specific category.	None (public)	None	200 OK - array of results ordered by position; 404 Not Found - category does not exist
POST	/api/enrolments/{id}/results	Captures or updates the finish time and position for an enrolment.	Organiser	{ finishTime, position, status }	200 OK - saved result; 403 Forbidden - not the owning organiser; 404 Not Found

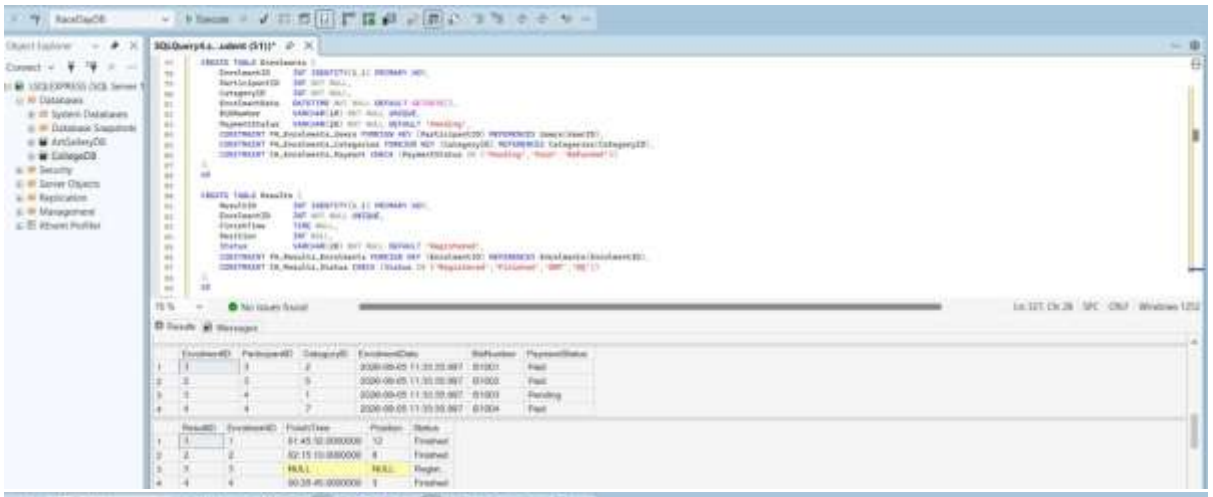
SECTION C

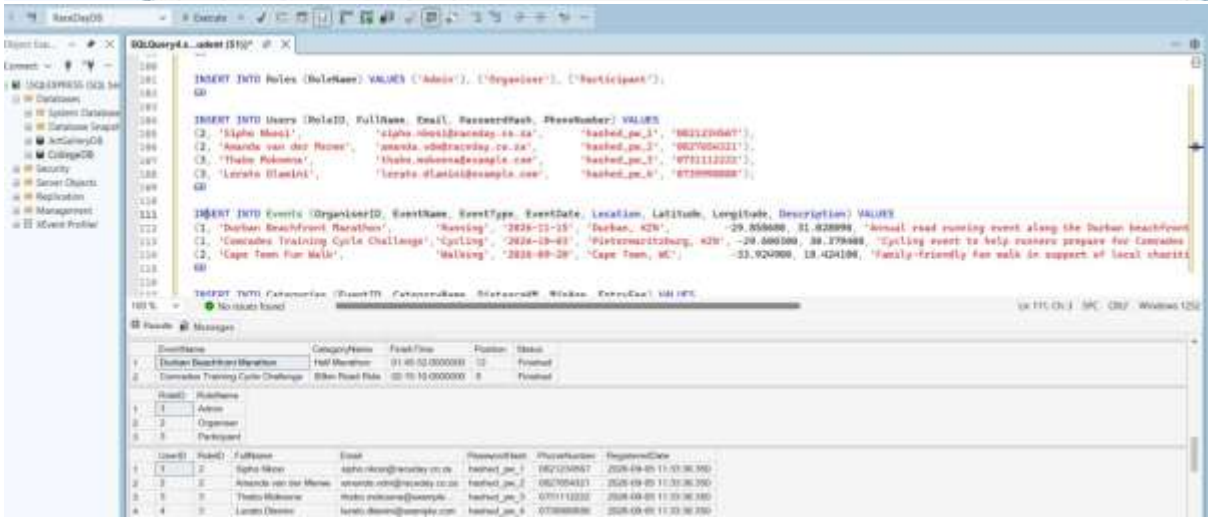


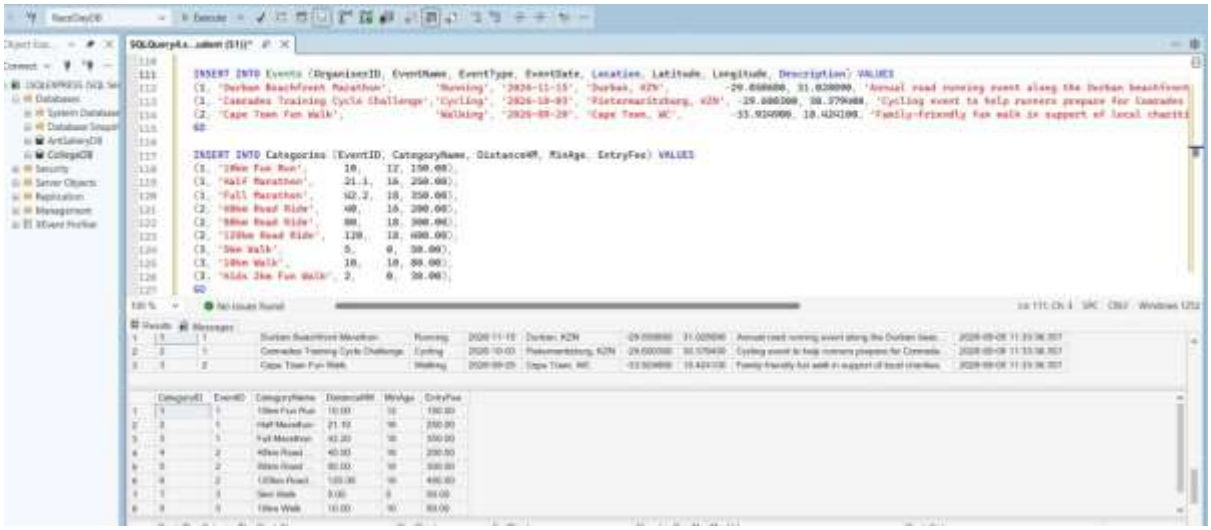

CREATE TABLE Roles (
RoleID INT IDENTITY(1,1) PRIMARY KEY,
RoleName VARCHAR(20) NOT NULL UNIQUE
);
GO
CREATE TABLE Users (
UserID INT IDENTITY(1,1) PRIMARY KEY,
RoleID INT NOT NULL,
FullName VARCHAR(100) NOT NULL,
Email VARCHAR(100) NOT NULL UNIQUE,
PasswordHash VARCHAR(255) NOT NULL,
PhoneNumber VARCHAR(20) NULL,
RegisteredDate DATETIME NOT NULL DEFAULT GETDATE(),
CONSTRAINT FK_Users_Roles FOREIGN KEY (RoleID) REFERENCES Roles(RoleID)
);
GO

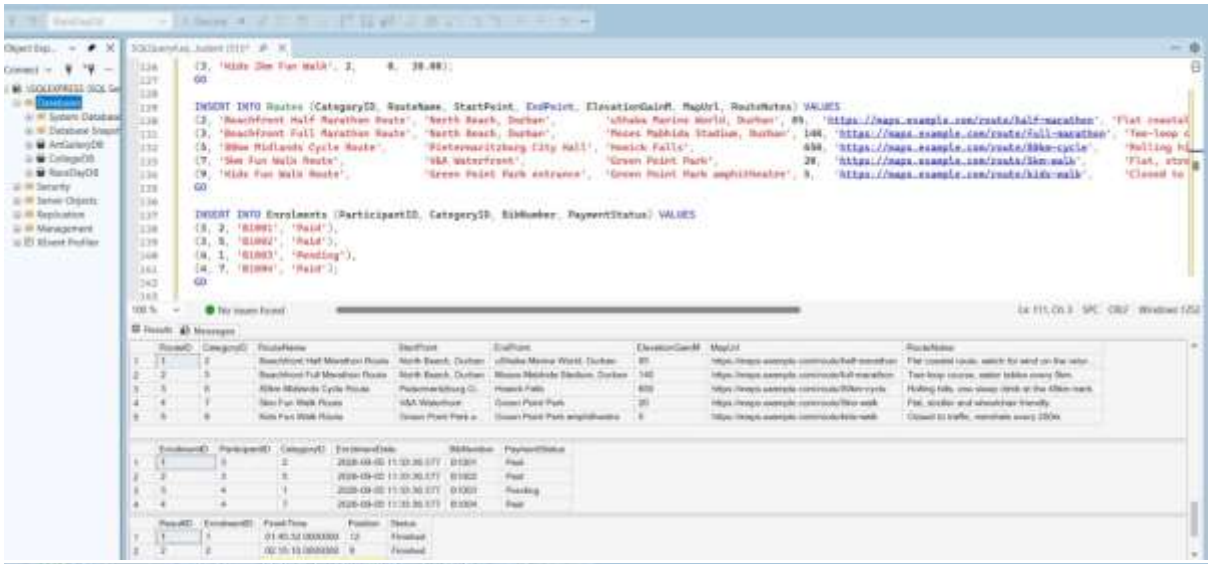

GO
CREATE TABLE Events (
EventID INT IDENTITY(1,1) PRIMARY KEY,
OrganiserID INT NOT NULL,
EventName VARCHAR(150) NOT NULL,
EventType VARCHAR(20) NOT NULL,
EventDate DATE NOT NULL,
Location VARCHAR(150) NOT NULL,
Latitude DECIMAL(9,6) NULL,
Longitude DECIMAL(9,6) NULL,
Description VARCHAR(500) NULL,
CreatedDate DATETIME NOT NULL DEFAULT GETDATE(),
CONSTRAINT FK_Events_Users FOREIGN KEY (OrganiserID) REFERENCES Users(UserID),
CONSTRAINT CK_Events_Type CHECK (EventType IN ('Running','Cycling','Walking'))
);
GO

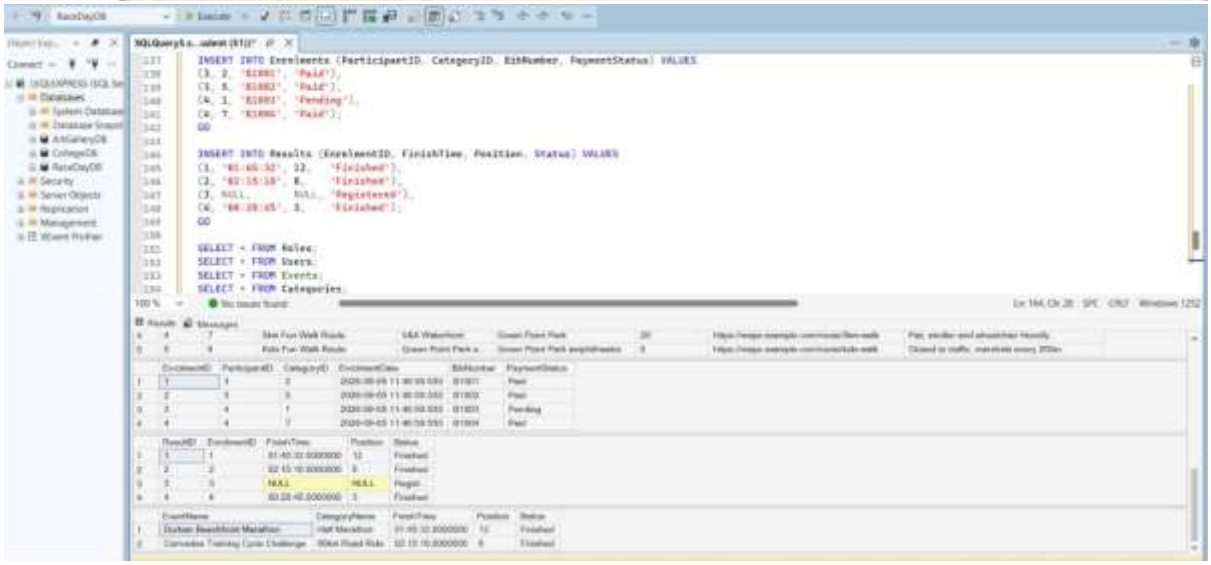


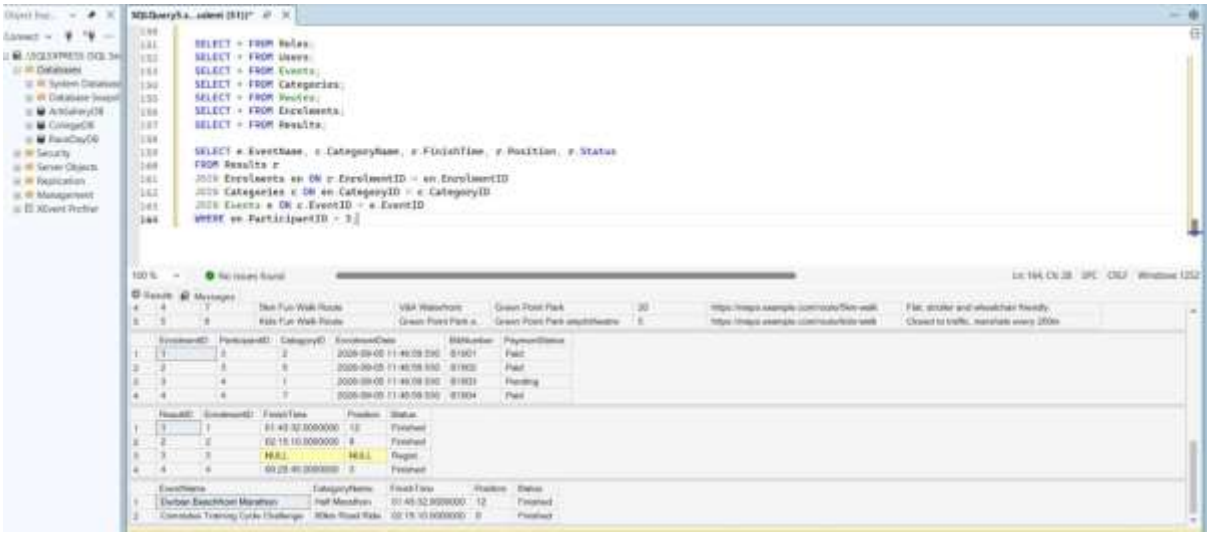












In section c the RaceDay schema SQL script builds full Race Day database in the SQL server by creating 7 tables from the ERD which are the Roles, Events, Categories, Routes, Enrolment and results. It has primary keys and foreign keys and constrains matching the ERD. The screenshots show the results of the SQL that illustrates the rows and tables.